IDEA SUBMISSION

J.A.R.V.I.S.

Justified Analysis, Reasoning & Visualization of Incident Systems

Autonomous Incident Investigation and Self-Healing SRE Platform

NITSHACKS 9.0 | AquaNox

Theme: Artificial Intelligence, Intelligent Systems & Distributed Systems

01 / PROPOSED SOLUTION

J.A.R.V.I.S. - Proposed Solution

Distributed systems produce fragmented alerts. J.A.R.V.I.S. turns those signals into one explainable incident, verifies recovery, and preserves an auditable post-mortem.

AI-POWERED RCA

- Multi-agent diagnosis with local Qwen support.
- Deterministic fallback rules keep investigations useful without an LLM.
- Competing hypotheses are recorded and falsified.

INTELLIGENT CORRELATION

- 120-second sliding window.
- Trace IDs, topology distance, shared dependencies, and causal signatures.
- Prevents unrelated failures from being merged.

SAFE SELF-HEALING

- Vetted runbooks: restart, pool recovery, stuck-job cleanup, rollback.
- Governance gate for risky actions.
- Synthetic health probes verify convergence.

02 / BUILD PLAN

TECHNICAL APPROACH

INGEST

- REST events and event batches
- AlertManager webhooks
- LOG/TXT/JSON/JSONL uploads
- Simulation and live telemetry

REASON

- FastAPI + Pydantic normalized schema
- SQLAlchemy SQLite/PostgreSQL storage
- Graph-based service topology
- LangGraph/LangChain agent workflow

ACT & EXPLAIN

- Next.js SRE cockpit
- WebSocket telemetry with REST fallback
- Timeline and correlated-event views
- Markdown post-mortem download

Flow: Multi-source ingestion → normalization → correlation → RCA/impact → governed remediation → verification → report

03 / REALITY CHECK

FEASIBILITY AND VIABILITY

FEASIBILITY

- Open-source stack and local AI reduce infrastructure cost.
- Mock microservices provide repeatable fault injection.
- SQLite enables local demos; PostgreSQL supports deployment.
- Modular APIs allow future observability integrations.

CHALLENGES & RISKS

- LLM availability and response latency.
- False positives in noisy telemetry.
- Automated actions can increase blast radius.
- Production credentials and permissions require strict control.

MITIGATION

- Deterministic fallbacks and explicit evidence.
- Topology + trace-aware correlation thresholds.
- Risk classification and human approval gates.
- Health verification and bounded runbooks.

04 / WHY IT MATTERS

IMPACT AND BENEFITS

TARGET AUDIENCE

- SRE and DevOps teams
- Cloud and infrastructure engineers
- Software development and operations teams
- Incident response and IT support teams

EXPECTED IMPACT

- Faster investigation and recovery
- Reduced duplicate-alert fatigue
- Clearer root-cause and blast-radius visibility
- Consistent timelines and post-mortems

DEMONSTRATED CAPABILITY

- Payment cascade scenario across four services
- Quantified failed transactions and logins
- Multi-source master-incident grouping
- Downloadable nine-section audit report

05 / EVIDENCE

RESEARCH AND REFERENCES

PROJECT EVIDENCE

- Repository: github.com/Rahul3106/Jarvis
- README architecture and API reference
- test_logs/ multi-source incident fixtures
- backend/tests/ phase-based verification suite

ENGINEERING FOUNDATION

- Google SRE Book: Monitoring Distributed Systems
- Distributed-systems incident response practices
- Graph-based dependency analysis
- Human-in-the-loop operational governance

CURRENT IMPLEMENTATION

- FastAPI ingestion and incident APIs
- Correlation engine with topology and traces
- Next.js interactive dashboard
- Agent workflow, remediation tools, timeline, and reports

J.A.R.V.I.S. converts fragmented telemetry into justified, verifiable operational decisions.